InspectraAI

THE PROBLEM

Industrial inspection is still a clipboard. The failure mode isn't carelessness — it's that nobody can answer **"has this actually changed since March?"** without standing in the same spot, in the same light, with the same memory. The hard part was never detecting rust; it is **change over time under uncontrolled capture conditions**. And when a system can't tell, it should stop and ask a human rather than guess — which is why ours does.

HOW ONE UPLOAD BECOMES A WORK ORDER

01 — Quality gate
Blur as variance of the Laplacian, plus exposure, glare, coverage. Below threshold → **pause**.

02 — Registration
ORB + `findHomography(RANSAC)`, scored on measured inlier ratio and reprojection error, warped with a validity mask.

03 — Change detection
absdiff → threshold → morphology → contours, cross-checked against SSIM. Regions get relative bboxes.

04 — Evidence → ticket
Six artifacts per finding, a finding with measured severity, and a priority-ranked maintenance ticket.

AGENTIC — MEASURED RESULTS DRIVE ACTIONS

WHAT THE VISION LAYER MEASURED	WHAT THE SYSTEM THEN DID
blur `0.4`, underexposed, coverage `99.7%`	**Run paused and asked the human. Resumed from its persisted context snapshot — not restarted.**
inlier ratio fell to `6.5%` after 3 registration attempts	**Paused a second time with a different, specific question quoting the measured reason.**
1 region, SSIM `0.971`, mean diff `2.4`	**Finding written with the measured bbox and severity derived from it.**
severity ≥ warning on safety-relevant equipment	**Maintenance ticket raised at high priority, awaiting approval.**
blur `1529.0` · `1612` keypoints · `99.9%` inliers · `2.41%` changed	**A 14-step trace persisted, each row with its own measured values and `duration_ms`.**

The planner is **a deterministic bounded state machine** — a transition table over states with three hard limits (iterations, wall-clock, repeat guard). `planner: "deterministic-rule-v1"` is recorded on every run. No language model is involved, and the app publishes that itself.

BENCHMARKS WE CAN DEFEND

Labelled synthetic corpus `32 pairs · seed 7`

IOU T	PREC	REC	F1	MIOU
0.30	0.8857	0.6889	0.775	0.572
0.50	0.4857	0.3778	0.425	0.732

$$IoU(B_p, B_g) = |B_p \cap B_g| \,/\, |B_p \cup B_g| \geq \tau$$

The same run "contradicting" its own headline was this: a looser τ accepts sloppier boxes, so recall rises while precision and mean IoU of survivors fall. **The UI labels the threshold and seed on screen** — a metric without them is unattributable.

WHAT WE FIXED, AND WHAT WE REFUSE TO CLAIM

- Our test fixture used `warpAffine(BORDER_REPLICATE)`, which **manufactured an edge band** the detector reported as whole-frame defects at **0.96 confidence on clean captures**. Found only by running 18 captures back to back — no unit test reached it.
- Alignment never wrote its validity mask, so the warp border scored as change. Fixed: **36 clean pairs now produce 0 false regions.**
- Correcting both **lowered** the headline score (0.90/1.00 → 0.886/0.689). That was the honest direction.
- **No AWS, S3 or COOL integration.** Runs on a Raspberry Pi 5. GET `/api/system/capabilities` publishes this — we shipped the endpoint that can refute us.
- Benchmark metrics describe a **synthetic** corpus, not real plant imagery.
- Video is frame sampling with measured sharpest-frame selection — **no per-frame defect tracking**, no live camera.

Stack · Next.js 15 · React 19 · TypeScript · Tailwind · Framer Motion · TanStack Query · FastAPI · SQLAlchemy 2 · Alembic · Pydantic v2 · OpenCV 5.0.0 · NumPy · Pillow · PostgreSQL 18 · Docker · Coolify · Traefik · Raspberry Pi 5 · BHASHINI NMT (12 languages, offline-cached)

LIVE DEMO
inspectraai.soumwadeepguha.com

SOURCE
github.com/soumwadeep/InspectraAI

VERIFY OUR CLAIMS
/api/system/capabilities

InspectraAI — System Architecture

One upload → measured change → evidence → finding → maintenance ticket, with a persisted trace of every decision

request / data · planner loop · not built

CLIENT · BROWSER

Next.js 15 · React 19

App Router dashboard. TypeScript strict, Tailwind design tokens, Framer Motion.

`TanStack Query — server state`
`TanStack Table — 15 data grids`

LanguageProvider

Translates rendered text via the server proxy. English source kept per node so the language can be reverted exactly.

`localStorage → offline replay`

Service Worker

App shell precache · stale-if-error on API reads · responses marked *served from cache* with their age.

Writes never queued

Offline means reads survive. An upload while disconnected fails loudly rather than pretending to save.

EDGE · SAME ORIGIN

Traefik

TLS + routing, managed by Coolify.

Next rewrites proxy

The browser only ever talks to one origin, so **CORS is not a deployment requirement**, and one build runs anywhere.

`/backend-api/:path*`
`→ BACKEND_ORIGIN/:path*`
`resolved at request time`

Auth boundary

JWT bearer → user → **organization id**. Every query is tenant-scoped; a cross-tenant id answers 404, not another tenant's row.

`get_current_user`
`get_current_organization_id`

Route guard

Unauthenticated visits render nothing until the session resolves — no protected chrome leaks for a frame.

APPLICATION SERVER · FASTAPI + OPENCV 5.0.0

One shared run service — `services/inspection_run.py`

The API endpoint and the demo seeder call the **same function**, so seeded data can never be structurally better than real data. Every dashboard number describes something the pipeline actually measured.

Vision pipeline — measured at every stage

01 Quality gate
`blur = ∇²I variance`
`exposure · glare`
`coverage · suitability`

02 Registration
`ORB + homography`
`RANSAC · inlier ratio`
`reprojection error`

03 Change detect
`absdiff → threshold`
`→ morphology → contours`
`cross-checked by SSIM`

04 Evidence
`6 artifacts / finding`
`annotated · crop · mask`
`comparison · overlay`

Deterministic bounded planner — not a language model

A transition table over states, choosing the next action from the measurement the last stage returned. Three hard limits stop it: iteration count, wall-clock budget, and a repeat guard. `planner: "deterministic-rule-v1"` is recorded on every run.

Human-in-the-loop — the run stops and asks

A failed quality gate or a registration that measures a near-zero inlier ratio pauses the run, **serialises its context to JSON**, and resumes from that snapshot when a human answers. It does not restart, and it does not guess.

Finding
`measured bbox`
`severity · confidence`

Maintenance ticket
`priority-ranked`
`guarded state transitions`

Persisted trace
`agent_steps`
`agent_tool_calls`

PERSISTENCE & EXTERNAL

PostgreSQL 18

23 tables, migrated with Alembic on container start. Enum **values** persisted, not Python names.

`organizations · equipment · inspections`
`agent_runs · agent_steps · tool_calls`
`findings · evidence · tickets ·`
`observations`

Filesystem evidence store

Derived PNGs written at runtime, served back over an **authenticated, tenant-scoped** route with path containment enforced.

BHASHINI NMT

12 Indic languages. The credential lives **only** on the server — a browser-side key would be published, not configured.

`POST /api/translate (batched)`

Self-auditing endpoint

Runtime introspection publishes what is real **and what is not** — including `language_model_planner: false`.

`GET /api/system/capabilities`

DESIGNED FOR · NOT IMPLEMENTED

AWS vision worker
EC2 / SageMaker heavy stage

S3 evidence storage
no boto3, no bucket

COOL library
AWS Marketplace · not used

Per-frame defect tracking
video = sampled frames

Runs today on a **Raspberry Pi 5** with local storage. **76** backend tests pass. These gaps are stated by the app itself, not hidden.